

MediUnify Healthcare Platform

SYSTEM ARCHITECTURE SPECIFICATION
VERSION 2.0 • CONFIDENTIAL

Enterprise Admin & Super Admin System Architecture & 1:1 Database Mapping

Platform: MediUnify Patient Ecosystem **Author:** Lead Enterprise Architect **Database Engine:** PostgreSQL / MongoDB (Mapped 1:1) **Date:** September 2026

Executive Intent: This document outlines the full multi-tier administrative governance model and the **exact field-level database schema** ensuring that patient-side app fields, backend server storage, and admin portals use **100% identical naming conventions** across all 14 clinical and operational modules.

1. EXPANDED 16-TIER ADMINISTRATIVE ROLES & ACCESS CONTROL (RBAC)

To operate a complex multi-vendor healthcare ecosystem spanning doctors, diagnostic laboratories, imaging centers, retail pharmacies, fertility clinics, home healthcare agencies, and emergency ambulances, MediUnify implements 16 discrete administrative roles:

Admin Role	Role Category	Primary Operational Scope	Authorized Modules	Audit & Compliance
SUPER_ADMIN	Master Controller	Unrestricted root access. System configs, financial payouts, global analytics, RBAC role grants, security overrides.	All Modules (1 to 14)	Full Audit Logging
CITY_OPS_ADMIN	Regional Operations	Manages specific city operations (Mysuru, Bengaluru, etc.). Provider SLAs, local dispatch zones, regional escalations.	Regional Clinics, Labs, Pharmacies	Regional Audit
CHIEF_MEDICAL_OFFICER	Clinical Governance	Doctor credentialing, medical license validation (KMC/MCI), clinical safety reviews, antibiotic prescribing audits.	Doctors, Video Consults, EHR	Medical Legal Log
HOSPITAL_CLINIC_ADMIN	Facility Network	Onboards partner hospitals (e.g., Apollo BGS, Unnathi Clinic). Surgery cost estimates, inpatient bed allocations.	Hospital Care, Surgeries, Quotes	Facility Audit
DOCTOR_PORTAL_ADMIN	Doctor Desk	Doctor slot calendars, leave management, in-clinic queuing, video telemedicine consultation room monitoring.	Appointments, Slots, Telemedicine	Doctor Activity Log
PATHOLOGY_LAB_ADMIN	Diagnostic Lab	Pathology test catalog (CBC, HbA1c, Thyroid), wellness packages, phlebotomist sample collection tracking, report sign-off.	Lab Tests, Packages, Reports	NABL Sign-off Log
PHLEBOTOMY_DISPATCHER	Home Collection	Assigns home sample collection requests to field phlebotomists. Barcode tracking, sample cold-chain temperature logs.	Home Samples, Phlebotomists	Sample Custody Chain
RADIOLOGY_CENTER_ADMIN	Imaging & Scans	MRI/CT/Ultrasound slot allocations across Mysore Scans, Bharath Diagnostics. Radiologist report uploads, DICOM archives.	Radiology Labs, Modalities, DICOM	Radiology Review Log
PHARMACY_STORE_ADMIN	Pharmacy & Inventory	Retail pharmacy inventory, medicine batch numbers, expiry dates, stock alerts, multi-store order routing.	Medicines, Inventory, Batches	Drug Controller Log
RX_VERIFICATION_PHARMACIST	Prescription Desk	Split-screen prescription review. Validates Schedule H/H1 drugs, approves generic substitutes, flags invalid prescriptions.	Prescription Queue, Order Approval	Pharmacist Sign-off
FERTILITY_IVF_COORDINATOR	IVF & Reproductive Care	Manages IVF/UI patient journeys, dedicated counselor assignments, ART consent forms vault, 0% EMI finance approvals.	Fertility, IVF/UI, ART Consents	ICMR ART Act Audit
HOME_CARE_SUPERVISOR	Home Healthcare	Verified GNM/ANM home nurses, daily patient vital check logs, wound dressing photo records, visit schedule audits.	Nurse Bookings, Staff Roster	Nurse Visit Logs
EQUIPMENT_RENTAL_ADMIN	Equipment Fleet	Hospital bed, oxygen concentrator, wheelchair inventory. Security deposits, delivery inspections, maintenance logs.	Equipment Rental, Deposits	Asset Inspection Log
EMERGENCY_SOS_DISPATCHER	Emergency Desk	24/7 live audio-visual emergency feed. Dispatches nearest BLS/ALS ambulance, pre-notifies destination casualty trauma team.	Ambulance SOS, GPS Dispatch	Emergency SLA Log
FINANCE_BILLING_ADMIN	Financials & Accounts	Doctor payout settlements, vendor commission splits, Care+ VIP memberships, wallet top-ups, refund approvals.	Ledgers, Payouts, Wallet, Gateway	Financial Ledger Audit

Admin Role	Role Category	Primary Operational Scope	Authorized Modules	Audit & Compliance
COMPLIANCE_PRIVACY_OFFICER	Governance & Legal	Audits every access to Electronic Health Records (EHR). Enforces DISHA, HIPAA, data privacy, and consent revocations.	Audit Trails, Access Logs, Security	Immutable Access Vault

2. DATABASE SCHEMA SPECIFICATION (100% IDENTICAL NAMING BETWEEN PATIENT APP, BACKEND & ADMIN)

To prevent data mismatches and broken synchronization, the table below specifies the **exact field names** used in the patient application (`database.json` , `dataSyncService.js` , `doctors.js` , `radiologyLabsData.js` , and `CartContext.js`) and their identical representation in the Super Admin database.

2.1 Patient & User Profile (`users` Collection / Table)

Database & App Field Name	Data Type	Patient App State Source	Super Admin Form / UI	Validation & Rules
<code>id</code>	String (PK)	<code>user.id</code> / <code>@userId</code>	Patient ID (Read-only)	Format: <code>PAT-xxxx</code>
<code>name</code>	String	<code>user.name</code> / <code>userName</code>	Full Patient Name Input	Required, min 2 chars
<code>email</code>	String	<code>user.email</code> / <code>userEmail</code>	Email Address	Valid Email Format, Unique
<code>phone</code>	String	<code>user.phone</code> / <code>userPhone</code>	Mobile Number	Normalized E.164 (+91)
<code>password</code>	String (Hashed)	<code>user.password</code>	Reset Password Input	bcrypt / argon2 hashed
<code>gender</code>	String	<code>user.gender</code>	Gender Selector Dropdown	'Male' 'Female' 'Other'
<code>age</code>	String	<code>user.age</code>	Age Field	e.g. '42 Yrs'
<code>city</code>	String	<code>user.city</code> / <code>selectedCity</code>	City Selector	Mysore, Bangalore, etc.
<code>bloodGroup</code>	String	<code>user.bloodGroup</code>	Blood Group Dropdown	O+, A+, B+, AB+, O-, etc.
<code>dob</code>	String	<code>user.dob</code>	Date of Birth Picker	Format: DD/MM/YYYY
<code>emergencyContact</code>	String	<code>user.emergencyContact</code>	Emergency Contact Info	e.g. '+91 98450 99999 (Brother)'
<code>address</code>	String	<code>user.address</code>	Street Address & Landmark	Complete residential address
<code>walletBalance</code>	Number (Decimal)	<code>user.walletBalance</code>	Wallet Balance (Credit/Debit)	Default 0, min 0
<code>carePoints</code>	Number (Integer)	<code>user.carePoints</code>	Care Points Balance	Loyalty points ledger
<code>isVipMember</code>	Boolean	<code>user.isVipMember</code>	CARE+ VIP Status Badge	True if active membership
<code>vipExpiryDate</code>	String (ISO Date)	<code>user.vipExpiryDate</code>	VIP Expiry Date	ISO 8601 Timestamp
<code>abhaId</code>	String	<code>user.abhaId</code>	ABHA Health Account ID	14-digit Ayushman Bharat ID

2.2 Family Members Sub-Document (`familyMembers` Array)

Database & App Field Name	Data Type	Patient App State Source	Super Admin Form / UI	Validation & Rules
<code>id</code>	String	<code>familyMember.id</code>	Member ID	Format: <code>FAM-xx</code>
<code>name</code>	String	<code>familyMember.name</code>	Full Name	Required
<code>relation</code>	String	<code>familyMember.relation</code>	Relationship Dropdown	Spouse, Son, Daughter, Father, etc.
<code>age</code>	String	<code>familyMember.age</code>	Age	Numerical string
<code>gender</code>	String	<code>familyMember.gender</code>	Gender	'Male' 'Female' 'Other'
<code>bloodGroup</code>	String	<code>familyMember.bloodGroup</code>	Blood Group	e.g. 'A+ Positive'
<code>phone</code>	String	<code>familyMember.phone</code>	Contact Phone	Optional or 10-digit mobile

2.3 Doctors & Specialists Master (`doctors` Collection / Table)

Database & App Field Name	Data Type	Patient App (<code>doctors.js</code>)	Super Admin Management	Rules & Notes
<code>id</code>	String (PK)	<code>item.id</code> (e.g. '1', '2')	Doctor ID	Unique identifier
<code>name</code>	String	<code>item.name</code>	Doctor Full Name	Prefix: 'Dr. '
<code>specialty</code>	String	<code>item.specialty</code>	Specialty Title	e.g. 'General Physician', 'Cardiologist'
<code>specialtyKey</code>	String	<code>item.specialtyKey</code>	Specialty Slug	'general', 'cardio', 'ortho', etc.
<code>qualification</code>	String	<code>item.qualification</code>	Medical Degrees	e.g. 'MBBS, MD (General Medicine)'
<code>experienceYears</code>	Number	<code>item.experienceYears</code>	Years of Clinical Practice	e.g. 12
<code>experience</code>	String	<code>item.experience</code>	Display Text	e.g. '12 Years Experience'
<code>rating</code>	Number	<code>item.rating</code>	Star Rating (0.0 to 5.0)	Computed from patient reviews
<code>reviewCount</code>	Number	<code>item.reviewCount</code>	Total Reviews Count	e.g. 380
<code>fee</code>	Number	<code>item.fee</code>	In-Clinic Consultation Fee (₹)	e.g. 500
<code>videoConsultFee</code>	Number	<code>item.videoConsultFee</code>	Video Consultation Fee (₹)	Configured per doctor
<code>clinicName</code>	String	<code>item.clinicName</code>	Affiliated Clinic / Hospital	e.g. 'Unnathi Multispecialty Clinic'
<code>clinicArea</code>	String	<code>item.clinicArea</code>	Area / Locality	e.g. 'Kuvempunagar, Mysore'
<code>clinicAddress</code>	String	<code>item.clinicAddress</code>	Full Physical Address	Includes pincode
<code>distanceKm</code>	Number	<code>item.distanceKm</code>	Distance Metric	Calculated from user GPS
<code>latitude</code>	Number	<code>item.latitude</code>	Map Latitude	e.g. 12.2858
<code>longitude</code>	Number	<code>item.longitude</code>	Map Longitude	e.g. 76.6341
<code>phone</code>	String	<code>item.phone</code>	Contact Phone	Clinic landline or mobile
<code>openHours</code>	String	<code>item.openHours</code>	Consultation Hours	e.g. '09:00 AM - 01:30 PM'
<code>availableToday</code>	Boolean	<code>item.availableToday</code>	Same-day Availability Flag	True / False toggle
<code>nextSlot</code>	String	<code>item.nextSlot</code>	Next Earliest Slot	e.g. 'Today, 04:30 PM'
<code>slots</code>	Array of Strings	<code>item.slots</code>	Available Time Slot Array	['09:30 AM', '11:00 AM', '04:30 PM']
<code>image</code>	String (URL)	<code>item.image</code>	Doctor Photo URL	High-res avatar URL
<code>about</code>	String	<code>item.about</code>	Doctor Bio & Clinical Focus	Detailed biography
<code>servicesOffered</code>	Array of Strings	<code>item.servicesOffered</code>	List of Specific Treatments	['General Consultation', 'Diabetes Care']
<code>badges</code>	Array of Strings	<code>item.badges</code>	Trust Badges	['Verified Doctor', 'Quick Consultation']

2.4 Central Appointments & Bookings (`appointments` Collection / Table)

Database & App Field Name	Data Type	Patient App (<code>database.json</code>)	Super Admin Oversight	Allowed Values
<code>id</code>	String (PK)	<code>appointment.id</code>	Appointment Booking Reference	Format: <code>APT-xxxx</code>
<code>userId</code>	String (FK)	<code>user.id</code>	Linked Patient Account	Foreign key to <code>users.id</code>
<code>patientName</code>	String	<code>appointment.patientName</code>	Patient Name at Booking	Self or Family Member
<code>serviceType</code>	String	<code>appointment.serviceType</code>	Service Category Filter	'doctor' 'video' 'nurse' 'lab' 'radiology'
<code>type</code>	String	<code>appointment.type</code>	Detailed Booking Type	'In-Person Consultation' 'Video Call'

Database & App Field Name	Data Type	Patient App (database.json)	Super Admin Oversight	Allowed Values
doctorId	String (FK)	doctor.id	Assigned Doctor ID	Foreign key to doctors.id
doctorName	String	appointment.doctorName	Doctor Display Name	e.g. 'Dr. Ramesh Kumar'
specialty	String	appointment.specialty	Doctor Specialty	e.g. 'Cardiology'
hospital	String	appointment.hospital	Hospital / Clinic Name	e.g. 'Apollo BGS Hospital, Mysore'
date	String	appointment.date	Appointment Date	Format: YYYY-MM-DD
time	String	appointment.time	Appointment Time	e.g. '10:30 AM'
timeSlot	String	appointment.timeSlot	Slot Range	e.g. '10:00 AM - 11:00 AM'
fee	Number	appointment.fee	Consultation Fee Charged (₹)	e.g. 650
status	String	appointment.status	Booking Lifecycle Status	'Scheduled' 'Confirmed' 'Completed' 'Cancelled'
meetingLink	String (URL)	appointment.meetingLink	Telemedicine Video Room Link	WebRTC / Agora meeting URL

2.5 Pharmacy Orders & Fulfillment (`orders` Collection / Table)

Database & App Field Name	Data Type	Patient App (<code>database.json</code>)	Super Admin & Pharmacy Desk	Allowed Values
<code>id</code>	String (PK)	<code>order.id</code>	Order ID	Format: <code>ORD-xxxx</code>
<code>userId</code>	String (FK)	<code>user.id</code>	Customer ID	Foreign key to <code>users.id</code>
<code>pharmacy</code>	String	<code>order.pharmacy</code>	Partner Retail Pharmacy Store	e.g. 'Apollo Pharmacy - Kuvempunagar'
<code>items</code>	Array of Objects	<code>order.items</code>	Line Items List	<code>[{ name, qty, price }]</code>
<code>total</code>	Number	<code>order.total</code>	Total Amount Paid (₹)	e.g. 210
<code>date</code>	String	<code>order.date</code>	Order Date	Format: YYYY-MM-DD
<code>status</code>	String	<code>order.status</code>	Fulfillment Stage	'Placed' 'RxApproved' 'OutForDelivery' 'Delivered' 'Cancelled'
<code>deliveryAddress</code>	String	<code>order.deliveryAddress</code>	Delivery Destination Address	Street, Area, Pincode
<code>deliveryPartnerName</code>	String	<code>order.deliveryPartnerName</code>	Assigned Delivery Driver	Driver Full Name
<code>deliveryPartnerPhone</code>	String	<code>order.deliveryPartnerPhone</code>	Driver Phone	Mobile Number
<code>otp</code>	String	<code>order.otp</code>	Delivery Verification OTP	4-digit secret OTP

2.6 Diagnostic Lab Tests & Bookings (`labBookings` Collection / Table)

Database & App Field Name	Data Type	Patient App Source	Lab Admin Management	Allowed Values
<code>id</code>	String (PK)	<code>labBooking.id</code>	Lab Booking ID	Format: <code>LAB-xxxx</code>
<code>userId</code>	String (FK)	<code>user.id</code>	Patient User ID	Foreign key to <code>users.id</code>
<code>patientName</code>	String	<code>labBooking.patientName</code>	Patient Name	Name of person tested
<code>tests</code>	Array of Objects	<code>labBooking.tests</code>	Diagnostic Tests List	CBC, Lipid Profile, Thyroid, etc.
<code>packageName</code>	String	<code>labBooking.packageName</code>	Package Title	e.g. 'Comprehensive Full Body Checkup'
<code>collectionType</code>	String	<code>labBooking.collectionType</code>	Sample Mode	'HomeSample' 'CenterWalkIn'
<code>date</code>	String	<code>labBooking.date</code>	Collection Date	Format: YYYY-MM-DD
<code>timeSlot</code>	String	<code>labBooking.timeSlot</code>	Sample Collection Slot	e.g. '07:00 AM - 08:30 AM'
<code>fee</code>	Number	<code>labBooking.fee</code>	Total Amount (₹)	Numerical INR
<code>status</code>	String	<code>labBooking.status</code>	Diagnostic Status	'Booked' 'SampleCollected' 'Testing' 'ReportReady'
<code>phlebotomistName</code>	String	<code>labBooking.phlebotomistName</code>	Assigned Phlebotomist	Staff Name
<code>reportUrl</code>	String (URL)	<code>labBooking.reportUrl</code>	Published PDF Report URL	Syncs directly to Health Vault

2.7 Radiology Scans & Imaging Centers (`radiologyBookings` Collection / Table)

Database & App Field Name	Data Type	Patient App (<code>radiologyLabsData.js</code>)	Radiology Admin Oversight	Allowed Values
<code>id</code>	String (PK)	<code>booking.id</code>	Scan Booking Reference	Format: <code>RAD-xxxx</code>
<code>userId</code>	String (FK)	<code>user.id</code>	Patient Account ID	Foreign key to <code>users.id</code>
<code>labId</code>	String (FK)	<code>lab.id</code>	Imaging Center ID	e.g. <code>lab-unnathi-main</code>
<code>labName</code>	String	<code>lab.name</code>	Center Display Name	e.g. 'Unnathi Advanced Diagnostics & 3T MRI'
<code>testId</code>	String	<code>test.id</code>	Scan Procedure Code	e.g. <code>RAD-UNN-01</code>

Database & App Field Name	Data Type	Patient App (radiologyLabsData.js)	Radiology Admin Oversight	Allowed Values
testName	String	test.name	Procedure Name	e.g. 'MRI Brain (Plain + Contrast)'
modalityCode	String	test.modalityCode	Machine Modality Type	'3T MRI' '128-Slice CT' '4D Ultrasound' 'Digital X-Ray'
date	String	booking.date	Scheduled Date	Format: YYYY-MM-DD
timeSlot	String	booking.timeSlot	Scan Slot Time	e.g. '11:00 AM - 11:45 AM'
fee	Number	test.price	Discounted Fee Paid (₹)	e.g. 4999
status	String	booking.status	Radiology Status	'Booked' 'SlotConfirmed' 'ScanDone' 'ReportAvailable'
reportUrl	String (URL)	booking.reportUrl	Official Radiologist PDF	Signed PDF Report URL

2.8 Home Nursing, Medical Equipment, IVF & Emergency Records

Module / Collection	Identical Key Names	Data Types	Patient App Reference	Admin Desk Handling
nurseBookings	id, userId, patientName, serviceName, nurseName, plan, startDate, endDate, timeSlot, address, status, fee	String, Number	database.json -> appointments (serviceType: nurse)	Home Care Supervisor manages staff nurse allocation, patient vitals, and visit reviews.
equipmentRentals	id, userId, patientName, equipmentName, dailyRate, durationDays, depositAmount, deliveryAddress, status	String, Number	EquipmentRentalScreen.js	Equipment Vendor Admin tracks bed/oxygen concentrator stock, security deposit, and pickup.
fertilityJourneys	id, userId, coupleName, clinicName, treatmentType, currentStage, coordinatorName, emiPlan, consentFormUrls	String, Array	FertilityIvfScreen.js, IVFJourneyScreen.js	Fertility Coordinator tracks IVF cycle steps (Stimulation, Retrieval, Transfer), consents, and EMI loans.
ambulanceDispatches	id, userId, patientName, patientPhone, callerLocation, ambulanceId, driverName, driverPhone, hospital, status	String, Object	AmbulanceScreen.js, EmergencyModal	24/7 SOS Dispatcher handles live audio alarm, sends nearest ambulance, notifies destination ER.
walletTransactions	id, userId, title, type, amount, date, status, gatewayRef	String, Number	database.json -> walletTransactions	Finance Admin reviews wallet top-ups, promotional bonuses, cashbacks, and refund disputes.

2.9 Concrete JSON Payload Examples (Exact Key-Value Pairs Stored in Database)

These JSON objects represent the exact format expected by both the Patient App (via `dataSyncService.js` / `database.json`) and the Super Admin Portal:

A. Patient Profile Record (users)

```
{
  "id": "PAT-1001",
  "name": "Rajesh Sharma",
  "email": "rajesh.sharma@example.com",
  "phone": "+91 98450 12345",
  "password": "$2b$10$hashedpass...",
  "gender": "Male",
  "age": "42 Yrs",
  "city": "Mysore",
  "bloodGroup": "O+ Positive",
  "dob": "15/08/1984",
  "emergencyContact": "+91 98450 99999 (Brother)",
  "address": "45, 4th Main, Kuvempunagar, Mysore",
  "walletBalance": 1250,
  "carePoints": 500,
  "isVipMember": true,
  "vipExpiryDate": "2027-09-19T00:00:00Z",
  "abhaId": "14-9845-0123-4567"
}
```

B. Doctor Master Record (doctors)

```
{
  "id": "1",
  "name": "Dr. Ananya Rao",
  "specialty": "General Physician",
  "specialtyKey": "general",
  "qualification": "MBBS, MD (General Medicine)",
  "experienceYears": 12,
  "experience": "12 Years Experience",
  "rating": 4.9,
  "reviewCount": 380,
  "fee": 500,
  "videoConsultFee": 400,
  "clinicName": "Unnathi Multispeciality Clinic",
  "clinicArea": "Kuvempunagar, Mysore",
  "clinicAddress": "No. 24, 5th Cross, Mysore - 570023",
  "distanceKm": 0.8,
  "latitude": 12.2858,
  "longitude": 76.6341,
  "phone": "+91 821 245 9901",
  "openHours": "09:00 AM - 01:30 PM",
  "availableToday": true,
  "nextSlot": "Today, 04:30 PM",
  "slots": ["09:30 AM", "11:00 AM", "04:30 PM"],
  "image": "https://images.unsplash.com/...",
  "about": "Renowned consultant physician..."
}
```

C. Appointment Record (appointments)

```
{
  "id": "APT-9001",
  "userId": "PAT-1001",
  "patientName": "Rajesh Sharma",
  "serviceType": "doctor",
  "type": "In-Person Consultation",
  "doctorId": "1",
  "doctorName": "Dr. Ramesh Kumar",
  "specialty": "Cardiology",
  "hospital": "Apollo BGS Hospital, Mysore",
  "date": "2026-09-12",
  "time": "10:30 AM",
  "timeSlot": "10:00 AM - 11:00 AM",
  "fee": 650,
  "paymentMethod": "wallet",
  "paymentStatus": "Paid",
  "status": "Confirmed",
  "meetingLink": "https://video.mediunify.com/APT-9001"
}
```

D. Pharmacy Order Record (orders)

```
{
  "id": "ORD-5501",
  "userId": "PAT-1001",
  "patientName": "Rajesh Sharma",
  "pharmacy": "Apollo Pharmacy - Kuvempunagar",
  "pharmacyId": "PHARM-MYS-01",
  "items": [
    { "id": "MED-01", "name": "Paracetamol 650mg", "qty": 2, "price": 45 },
    { "id": "MED-02", "name": "Vitamin C 500mg", "qty": 1, "price": 120 }
  ],
  "total": 210,
  "discount": 0,
  "deliveryFee": 0,
  "date": "2026-09-07",
  "deliveryAddress": "45, 4th Main, Kuvempunagar",
  "status": "Delivered",
  "deliveryPartnerName": "Suresh Kumar",
  "otp": "4821"
}
```

3. END-TO-END OPERATIONAL WORKFLOWS (PATIENT APP → DATABASE → ADMIN)

1. Doctor In-Clinic / Video Booking

Patient books slot in app → Saved to `appointments` with status `'Confirmed'` → Doctor Admin sees live appointment queue → Video room link generated → Completed → Rx uploaded.

2. Doorstep Pharmacy Order

Patient adds medicines to cart → Uploads Rx → Rx Pharmacist verifies matching drugs → Routed to closest partner pharmacy → Delivery agent dispatched with OTP → Status marked `'Delivered'`.

3. Diagnostic Pathology Sample

Patient selects test package → Chooses morning slot → Phlebotomy Dispatcher assigns field phlebotomist → Barcode sample scanned → Lab testing → Pathologist signs off PDF → Syncs to patient vault.

4. Emergency SOS Dispatch

Patient taps "Ambulance SOS" → High-priority WebSocket alert triggers on Admin console → Nearest ALS/BLS ambulance dispatched → Hospital casualty pre-alerted with bed pre-booking.

4. TECHNICAL ARCHITECTURE, SECURITY & IMPLEMENTATION STACK

- Data Integrity Principle:** The Admin Portal and Patient App must never use diverging schemas. Any column added to the patient app must be mirrored in the Super Admin system using the identical camelCase name.
- Admin Portal Framework:** Next.js 14 / React 19 web application hosted on secure sub-domain (e.g., `admin.mediunify.com`).
 - API Gateway & Authentication:** Node.js / Express backend with JWT tokens, refresh tokens, role-based middleware guards, and mandatory 2FA (SMS OTP or TOTP Authenticator).
 - Audit Trail Logger:** Immutable append-only audit ledger recording every read, write, or export action performed on patient medical histories in compliance with DISHA (Digital Information Security in Healthcare Act).
 - Real-time Layer:** WebSockets (Socket.io) powering live ambulance GPS movement, urgent video call requests, and real-time pharmacy delivery statuses.

